



GHULAM ISHAQ KHAN INSTITUTE • DATA VISUALIZATION • 2026

AQI Scale: Good ▶ Fair ▶ Moderate ▶ Poor ▶ Very Poor

Air Quality Narrative
Project Final — Complete Report

Step 1: Data Analysis • Step 2: Storyboard • Step 3: Visual Design Study

Dataset:	concatenated_dataset_Aug_2021_to_July_2024.csv (Kaggle)
Records:	123,134 hourly observations 18 variables
Period:	August 2021 – June 2024 (3 years)
Research Q:	What drives poor air quality, and when is it worst?
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Step 1: Data Analysis & Exploration

This section documents the exploratory data analysis conducted on the air quality and meteorological dataset. The analysis guided the research question, shaped the narrative arc, and produced the insights presented in the final visualization.

1.1 Dataset Overview

Attribute	Detail
File	concatenated_dataset_Aug_2021_to_July_2024.csv
Source	Kaggle
Period	August 24, 2021 — June 30, 2024
Records	123,134 rows (hourly observations)
Variables	18 columns — 8 pollutant components + 7 weather variables + AQI + datetime
Missing	0 — dataset is completely clean
AQI Scale	1 (Good) → 2 (Fair) → 3 (Moderate) → 4 (Poor) → 5 (Very Poor)

1.2 Research Question & Motivation

Research Question What drives poor air quality over this 3-year period — and when (by season and by hour of day) is pollution at its worst?

This question was motivated by Pakistan’s well-documented annual smog crisis. Cities like Lahore and Peshawar regularly rank among the world’s most polluted. Personal experience of visible smog during winter months, combined with news coverage of school closures and health advisories, prompted us to ask whether the data confirms the perceived seasonal pattern — and what meteorological conditions make it worse. Sources consulted include the WHO Air Quality Report 2023, IQAir World Air Quality Report 2023, and Dawn News coverage of the Lahore smog seasons 2022–23.

1.3 Phase 1 — Data Structure & Quality

The following checks were performed to establish data quality and structure:

- All 18 columns verified for correct data types — datetime parsed, numerics confirmed
- Zero missing values across all 123,134 rows
- All pollutant distributions are right-skewed — extreme pollution events occur but are less frequent
- PM2.5 and PM10 are strongly correlated ($r \approx 0.9$) — shared combustion sources
- Temperature shows bimodal distribution — cold winters vs hot summers
- Precipitation near-zero for most hours — confirming arid baseline conditions

AQI Level	Label	Hours	Percentage
1	Good	1,689	1.4%
2	Fair	11,794	9.6%
3	Moderate	23,399	19.0%
4	Poor	26,887	21.8%
5	Very Poor	59,365	48.2%

Nearly 48% of all hours are at AQI 5 — the worst possible category. Over 85% of hours fall at AQI 4 or 5.

1.4 Phase 2 — Research Question Investigation

Q1: How does air quality vary across seasons and months?

Temporal features (month, season, hour, day-of-week) were extracted from the datetime column. Monthly and seasonal aggregations were computed.

Season	Months	Avg AQI	Finding
Winter	Dec–Feb	4.40	Worst season — near-permanent AQI 5 conditions
Autumn	Sep–Nov	4.35	Second worst — crop burning begins in Oct–Nov
Spring	Mar–May	3.75	Improving — warming temperatures aid dispersal
Summer	Jun–Aug	3.67	Best season — monsoon rains + strong winds

Follow-up finding: PM2.5, CO, NO, and NO₂ all peak in winter (combustion sources). Ozone (O₃) peaks in summer due to photochemical reactions with UV sunlight — a distinct mechanism. Year-over-year comparison confirms the pattern repeats consistently across all 3 years.

Q2: How do weather conditions influence air quality?

Pearson correlation coefficients were computed between all weather variables and AQI. Scatter plots, binned bar charts, and rainy vs dry day comparisons were performed.

Weather Variable	Correlation with AQI	Interpretation
Surface Pressure	$r = +0.318$	Strongest predictor — high pressure = temperature inversions
Precipitation	$r = -0.110$	Rain washes PM2.5 from the air via wet deposition
Wind Speed	$r = -0.105$	Wind disperses pollutants — calm air = worse AQI
Relative Humidity	$r = -0.052$	Minor negative effect on AQI
Temperature	$r = -0.021$	Weak direct effect; acts primarily via pressure/season

Key finding: Surface pressure is the dominant weather driver — high pressure systems create temperature inversions that trap pollutants near the ground. Rain and wind both offer temporary relief. Calm air (0–1 m/s) produces consistently the worst AQI readings.

Q3: Are there consistent daily hourly pollution patterns?

Hourly averages were computed across all 3 years. A radial clock chart, bar chart, and weekday vs weekend comparison were produced.

Time Window	Avg AQI	Reason
1–5 AM (Peak)	~4.6	Temperature inversions trap pollutants overnight
7–9 AM	~4.2	Morning rush hour traffic spike
12–3 PM (Best)	~3.9	Solar heating mixes atmosphere, disperses particles
6–9 PM	~4.1	Evening rush hour + cooling air begins

Weekday mornings (Mon–Fri, 7–9 AM) show measurably higher PM_{2.5} than the same hours on weekends — the fingerprint of commuter traffic. The afternoon window (1–3 PM) is consistently the safest time for outdoor activity.

1.5 Key Takeaways from EDA

- Winter (Dec–Feb) average AQI of 4.40 vs Summer 3.67 — a 20% difference
- 48.2% of all hours recorded at AQI 5 (Very Poor)
- High surface pressure is the single strongest meteorological predictor ($r = +0.318$)
- Rain and wind both measurably improve AQI in the short term
- 2–4 AM is the most polluted window; 1–3 PM is the safest
- Combustion pollutants (PM_{2.5}, CO, NO₂) dominate winter; O₃ peaks in summer

Step 2: Storyboard

The storyboard defines the complete narrative arc of the scrollytelling web visualization. Eight scenes progress from a dramatic hook through data exploration to actionable conclusions, combining author-driven and reader-driven approaches.

2.1 Narrative Arc

Beat	Section	Type	Purpose
1	Hook	Author-driven	Shocking stat: 48.2% of hours at AQI 5
2	AQI Context	Author-driven	What is AQI? Scale explanation with health advisories
3	3-Year Timeline	Author-driven	Animated area chart — the cyclical winter pattern
4	Seasonal Dive	Hybrid	Season cards + monthly bar chart — scroll to reveal
5	Weather Drivers	Hybrid	Correlation chart + 3 weather factor cards
6	Daily Cycle	Hybrid	Radial clock chart + hourly bar — 24h pollution rhythm
7	Free Explorer	Reader-driven	Year/metric filter — user-controlled exploration
8	Conclusion	Author-driven	3 takeaway cards + policy recommendations

2.2 Scene-by-Scene Storyboard

Scene 1 — The Hook — A City Choking

Description:	Full-screen dark background. A counter animates from 0% to 48.2%, followed by the label 'of all hours were at AQI 5 — Very Poor'. A pulsing purple radial ring emanates behind the number for visual urgency.
Visualization:	Static — pure author narrative. No interaction required.
Interaction:	Scroll-triggered counter animation. 3 stat chips below (123K records, 3 years, 18 variables). Arrow + 'Scroll to explore' prompt at bottom.
Annotation:	Animated counter counts up on page load to maximise impact.

Scene 2 — AQI Scale — Building Context

Description:	5 horizontal colored bands (AQI 1–5) with label, percentage, and health advisory. Stat cards below show total records, variables, time span, missing values.
Visualization:	Hover over each AQI band to reveal health advisory tooltip with record count.
Interaction:	Annotation box: 'Over 85% of hours fall at AQI 4 or 5 — safe air is the exception.' Each band shows the exact percentage from the dataset.
Annotation:	Bands animate in sequentially left to right on scroll-entry.

Scene 3 — 3-Year Timeline

Description:	Full-width animated area chart of daily average AQI from Aug 2021 to Jun 2024. Purple shaded winter bands highlight Nov–Feb each year. WHO guideline shown as dashed line at AQI 3.
Visualization:	Hover anywhere on chart to see exact date, AQI, and PM2.5 value in tooltip.
Interaction:	Text annotation: winter smog seasons labeled. WHO line labeled 'WHO Moderate'. Chart animates drawing left-to-right on scroll-entry.
Annotation:	Area draws itself with D3 stroke-dashoffset animation over 2 seconds.

Scene 4 — Seasonal Deep-Dive

Description:	Four season cards (Winter/Spring/Summer/Autumn) each showing average AQI as a large number with color coding. Below: monthly bar chart with season-colored bars.
Visualization:	Hover bars to see exact monthly average. Season cards highlight on hover.
Interaction:	‘▲ WORST’ badge on Winter card. Bar chart bars animate growing upward on entry. Annotation: ‘O ₃ peaks in summer — a different pollutant mechanism from winter PM2.5.’
Annotation:	Season cards and bar chart bars stagger-animate in sequence on scroll-entry.

Scene 5 — Weather Drivers

Description:	Horizontal bar chart showing Pearson correlation of all weather variables with AQI (teal = negative correlation, red = positive). Three detail cards below for the top 3 drivers: Surface Pressure, Precipitation, Wind Speed.
Visualization:	Hover bars to see exact correlation value and variable name.
Interaction:	Bar chart annotated with r values. Cards explain the physical mechanism in plain language. Bars animate from center outward on scroll-entry.
Annotation:	Color encodes direction: teal = helps (negative r), red = hurts (positive r).

Scene 6 — 24-Hour Daily Cycle

Description:	Radial clock chart: 24 segments around a circle, bar length = avg AQI for that hour. Color-coded by AQI level. Beside it: standard horizontal bar chart of same data with rush hours highlighted.
Visualization:	Hover any radial segment or bar to see hour, avg AQI, and avg PM2.5.
Interaction:	Annotations: ‘2 AM — Peak pollution’ and ‘2 PM — Safest hour’ and ‘7–9 AM — Rush hour spike’. Radial chart animates clockwise on scroll-entry.
Annotation:	Each segment grows outward sequentially, completing the clock face animation.

Scene 7 — Free Explorer

Description:	Interactive dashboard: Year dropdown (All/2021/2022/2023/2024) + Metric dropdown (AQI / PM2.5). Chart updates dynamically. Reset button. Tip text below guides user.
Visualization:	Full filter controls update chart in real time. Hover tooltip on chart. Reset button.
Interaction:	Default hint: 'Select 2022 and compare with 2023 to see year-over-year smog change.' Chart title updates to reflect current filter selection.
Annotation:	All chart transitions animate smoothly at 300ms for filter changes.

Scene 8 — Conclusion & Policy

Description:	Three conclusion cards stagger in: Q1 (Seasonal), Q2 (Weather), Q3 (Daily Cycle). Below: policy recommendations panel with 3 numbered action items derived from data. Footer with team names, registration numbers, and dataset acknowledgement.
Visualization:	Static — pure author narrative. Cards animate in with staggered delay.
Interaction:	Each card shows a bold headline stat. Policy items numbered with teal badges. Dataset source and course reference in footer.
Annotation:	Cards slide up and fade in sequentially with 300ms delay between each.

2.3 Navigation Model

- Progress bar: Fixed at top — shows scroll position through the full story (0–100%)
- 8 nav dots: Fixed on right side — click any dot to jump directly to that scene
- Scenes 1–3 and 8: Author-driven — content appears and animates on scroll
- Scenes 4–6: Hybrid — charts animate on scroll-entry, then become interactive
- Scene 7: Fully reader-driven — user controls all filters and exploration
- Inter-scene transitions: 400ms fade for narrative continuity
- Responsive layout: Single column on mobile, multi-column on desktop

Step 3: Visual Design Study

The visual design study establishes the look, feel, and tone of the narrative visualization. A diverse range of visual media was examined to inform decisions on color, typography, layout, and interaction design.

3.1 Inspirational References

References were drawn from data journalism, print, UI design, and books to ensure a diverse design vocabulary:

Source	Medium	Design Element Borrowed
NYT ‘The Air We Breathe’ (2023)	Web / Data Journalism	Dark background with AQI color scale; scrollytelling pacing and annotation style
IQAir World Air Quality Report	Print PDF / Infographic	AQI color gradient (green→purple); large stat callouts; minimal grid layouts
The Pudding — Visual Essays	Web	Full-viewport sections; smooth transitions; reader engagement hooks
Bloomberg CityLab Smog Coverage	Web / Newsprint	Annotated line charts; narrative text interspersed with data charts
NASA Earth Observatory	Web / Photography	Dark space-like aesthetic; high contrast; glowing data point effects
Tufte — Envisioning Information	Book	Data-ink ratio; eliminating chartjunk; small multiples for comparison
Figma Dark Dashboard UI Kits	UI Design	Card-based layouts; teal accent on navy; clean sans-serif hierarchy
National Geographic Climate	Magazine / Print	Dramatic color for urgency; annotations as primary storytelling devices

3.2 Color Palette

The palette is content-driven rather than decorative. Navy evokes nighttime and smog-laden atmosphere. Teal signals environmental/scientific data and provides high contrast on dark backgrounds. AQI level colors are preserved from WHO/IQAir international standards so readers immediately recognize the encoding:

Role	Name	Hex	Rationale
Primary BG	Deep Navy	#0A1628	Evokes night sky and smog; high contrast for data
Accent	Teal	#0D9488	Environmental + scientific; pops on dark backgrounds

Accent Light	Teal Light	#14B8A6	Hover states, secondary highlights
Surface	Mint Tint	#CCFBF1	Light card surfaces on interactive scenes
AQI 1	Good Green	#00C851	WHO/IQAir standard — internationally recognized
AQI 2	Fair Yellow	#FFCC00	WHO/IQAir standard — internationally recognized
AQI 3	Moderate Orange	#FF8800	WHO/IQAir standard — internationally recognized
AQI 4	Poor Red	#FF3300	WHO/IQAir standard — internationally recognized
AQI 5	Very Poor Purple	#8B1FA8	WHO/IQAir standard — internationally recognized
Body Text	Slate	#1E293B	Near-black for maximum readability on light backgrounds
Muted Text	Gray	#64748B	Captions, labels, secondary information

3.3 Typography

Inter was chosen as the sole typeface. It is a clean, modern, open-source sans-serif designed specifically for screen readability. Its geometric construction reinforces the analytical, data-driven tone without decorative distraction. Inter Mono is used for numerical chart labels to ensure consistent digit alignment.

Element	Font	Size	Weight	Purpose
Hook Statistic	Inter	4–6rem	Black 900	Commands immediate attention for the 48% hook
Section Title	Inter	1.8rem	Bold 700	Clear hierarchy; readable at scroll speed
Body Narrative	Inter	1rem	Regular 400	Neutral — keeps focus on data, not decoration
Chart Labels	Inter Mono	0.75rem	Regular	Monospace ensures consistent number alignment
Annotations	Inter	0.8rem	Medium Italic	Italic signals interpretive voice vs data labels
AQI Labels	Inter	0.7rem	Bold CAPS	All-caps for categorical distinction

3.4 Layout & Composition Principles

Full-viewport sections: Each scene occupies 100vh. One idea per screen prevents cognitive overload and enforces scrollytelling rhythm.

Data-ink ratio (Tufte): Non-essential grid lines, tick marks, and chart borders are removed. Color encodes meaning — never used decoratively.

Dark–light sandwich: Author-driven scenes (1,2,3,8) use dark navy for drama and urgency. Reader-driven scenes (4–7) use lighter mint/white for comfortable extended interaction.

Breathing room: Minimum 80px vertical padding between content blocks. Charts are never crowded against text. White space is a design element.

Hierarchy via size: Hook statistic at 5rem, titles at 1.8rem, body at 1rem. Size ratio communicates importance without requiring color alone.

Consistent teal motif: A thin teal left-border accent on all information cards creates visual continuity across all 8 scenes.

3.5 Interaction & Animation Design

All animations are functional — each serves a specific narrative or comprehension purpose, never purely decorative:

Technique	Scene	Purpose
Counter 0→48.2%	1	Creates suspense — forces reader engagement before reading
Chart draw (stroke-dashoffset)	3	Mirrors discovery — data appears as you scroll and explore
Staggered bar entry	4, 6	Guides eye to next data point; prevents visual overwhelm
Hover tooltips	2–7	Rewards curiosity; exact values without cluttering chart
Smooth filter transitions	7	Maintains data continuity when user changes filters
Radial clock animation	6	Metaphorically links chart type to time-of-day concept
Inter-scene fade	All	Signals narrative progression; reinforces brand identity
Conclusion card stagger	8	Builds to a climax — each finding lands separately

3.6 Accessibility

- All color encodings are supplemented with text labels — readable without color perception
- Contrast ratio meets WCAG AA minimum (4.5:1) on both dark and light backgrounds
- All interactive elements are keyboard-navigable
- Tooltip text shown as visible captions on mobile (no hover available)
- Font sizes in rem units — scales with user browser font preferences

3.7 Implementation & Deployment

The narrative visualization is implemented as a single self-contained HTML file using D3.js v7 and vanilla CSS/JS. Dataset values are pre-processed and embedded directly as JavaScript arrays — no server required. The file is deployed via GitHub Pages.

Technology	Role
D3.js v7	All data visualizations — area charts, bar charts, radial chart, correlation chart
Intersection Observer	Scroll-triggered animations and scene fade-in effects
Vanilla CSS	Dark theme, responsive grid, card components — no external CSS framework
GitHub Pages	Free static hosting directly from repository — public URL for submission
Pre-processed JSON	Dataset aggregated to daily/hourly/monthly/seasonal averages in JS

GitHub Pages Deployment URL

- https://murtaza436.github.io/DV_PROJECT/